2520090050 T SHANMUKHA SAGAR THATHA CNNP SKILL 3

```
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [
1195 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [14
50 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [
238 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components
 [46.4 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packag
es [1228 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe Translation
-en [248 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Compo
nents [76.3 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Pac
kages [1788 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [3
15 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components
[181 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted Translati
on-en [411 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Package
s [1711 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-e
n [344 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Compone
nts [388 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packa
ges [1883 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation
-en [430 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Compo
nents [940 B]
Get:22 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Component
s [5744 B]
Get:23 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Compo
nents [12.6 kB]
Fetched 12.3 MB in 40s (309 kB/s)
Reading package lists... Done
```

```
manchodu@Srini:~$ sudo apt install iperf3 iproute2 wireshark iputils-ping -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
iperf3 is already the newest version (3.16-1build2).
iproute2 is already the newest version (6.1.0-1ubuntu6.4).
iproute2 set to manually installed.
wireshark is already the newest version (4.2.2-1.1build3).
iputils-ping is already the newest version (3:20240117-1ubuntu0.1).
iputils-ping set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 144 not upgraded.
manchodu@Srini:~$
```

```
manchodu@Srini:~$ iperf3 --version
iperf 3.16 (cJSON 1.7.15)
Linux Srini 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun
 18 21:54:43 UTC 2026 x86_64
Optional features available: CPU affinity setting, IPv6 flow label, SCTP, TC
P congestion algorithm setting, sendfile / zerocopy, socket pacing, authenti
cation, bind to device, support IPv4 don't fragment, POSIX threads
manchodu@Srini:~$
```

```
manchodu@Srini:~$ tc -V
tc utility, iproute2-6.1.0, libbpf 1.3.0
manchodu@Srini:~$ wireshark --version
Wireshark 4.2.2 (Git v4.2.2 packaged as 4.2.2-1.1build3).

Copyright 1998-2024 Gerald Combs <gerald@wireshark.org> and contributors.
Licensed under the terms of the GNU General Public License (version 2 or later).
This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compiled (64-bit) using GCC 13.2.0, with GLib 2.80.0, with Qt 6.4.2, with
libpcap, with POSIX capabilities (Linux), with libnl 3, with zlib 1.3, with
PCRE2, with Lua 5.2.4, with GnuTLS 3.8.3 and PKCS #11 support, with Gcrypt
1.10.3, with Kerberos (MIT), with MaxMind, with nghttp2 1.59.0, with nghttp3
0.8.0, with brotli, with LZ4, with Zstandard, with Snappy, with libxml2 2.9.14,
with libsmi 0.4.8, with QtMultimedia, without automatic updates, with Minizip,
with binary plugins.

Running on Linux 6.18.33.2-microsoft-standard-WSL2, with 13th Gen Intel(R)
Core(TM) i5-13420H (with SSE4.2), with 7802 MB of physical memory, with GLib
2.80.0, with Qt 6.4.2, with libpcap 1.10.4 (with TPACKET_V3), with zlib 1.3,
with PCRE2 10.42 2022-12-11, with c-ares 1.27.0, with GnuTLS 3.8.3, with Gcrypt
1.10.3, with nghttp2 1.59.0, with nghttp3 0.8.0, with brotli 1.1.0, with LZ4
1.9.4, with Zstandard 1.5.5, with libsmi 0.4.8, with LC_TYPE=C.UTF-8, binary
plugins supported.
```

```
manchodu@Srini:~$ sudo ip addr add 192.16
manchodu@Srini:~$ sudo ip addr add 192.16
manchodu@Srini:~$ sudo ip addr add 192.16
Cannot find device "eth1"
manchodu@Srini:~$ sudo sysctl -w net.ipv4
net.ipv4.ip_forward = 1
manchodu@Srini:~$ sudo ip addr add 192.16
manchodu@Srini:~$ sudo ip route add 192.1
RTNETLINK answers: File exists
manchodu@Srini:~$ sudo ip route add 192.1
RTNETLINK answers: File exists
manchodu@Srini:~$ ping 192.168.20.2
PING 192.168.20.2 (192.168.20.2) 56(84) b
64 bytes from 192.168.20.2: icmp_seq=1 tt
```

```
manchodu@Srini:~$ sudo ip addr add 192.168.10.2/24 dev eth0
manchodu@Srini:~$ sudo ip addr add 192.168.10.1/24 dev eth0
manchodu@Srini:~$ sudo ip addr add 192.168.20.1/24 dev eth1
Cannot find device "eth1"
manchodu@Srini:~$ sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
manchodu@Srini:~$ sudo ip addr add 192.168.20.2/24 dev eth0
manchodu@Srini:~$ sudo ip route add 192.168.20.0/24 via 192.168.10.1
RTNETLINK answers: File exists
manchodu@Srini:~$ sudo ip route add 192.168.10.0/24 via 192.168.20.1
RTNETLINK answers: File exists
manchodu@Srini:~$ ping 192.168.20.2
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=64 time=0.034 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=64 time=0.036 ms
64 bytes from 192.168.20.2: icmp_seq=3 ttl=64 time=0.053 ms
64 bytes from 192.168.20.2: icmp_seq=4 ttl=64 time=0.047 ms
64 bytes from 192.168.20.2: icmp_seq=5 ttl=64 time=0.047 ms
64 bytes from 192.168.20.2: icmp_seq=6 ttl=64 time=0.258 ms
64 bytes from 192.168.20.2: icmp_seq=7 ttl=64 time=0.102 ms
64 bytes from 192.168.20.2: icmp_seq=8 ttl=64 time=0.087 ms
```

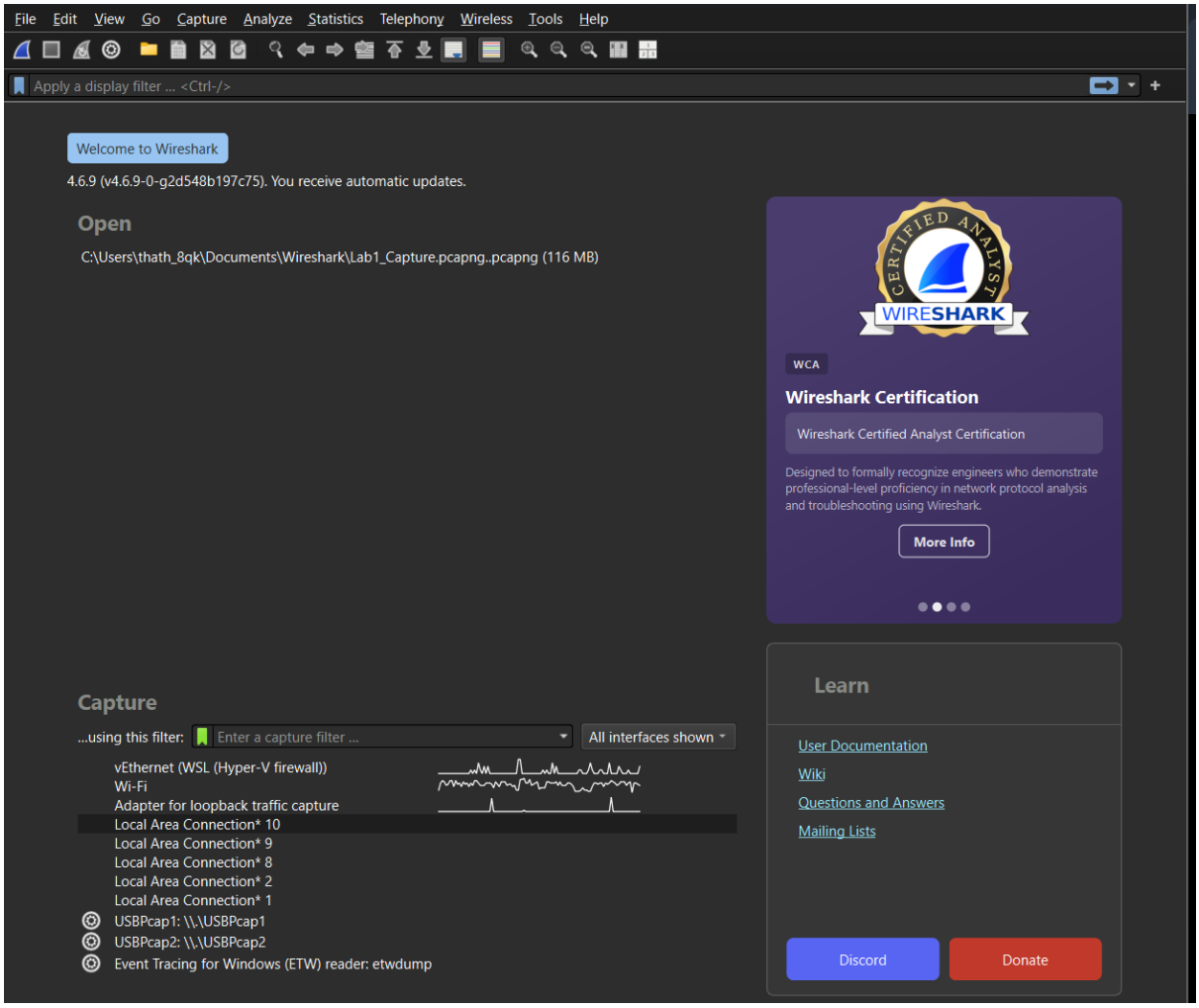
```
manchodu@Srini:~$ sudo ip addr add 192.168.10.2/24 dev eth0
manchodu@Srini:~$ sudo ip addr add 192.168.10.1/24 dev eth0
manchodu@Srini:~$ sudo ip addr add 192.168.20.1/24 dev eth1
Cannot find device "eth1"
manchodu@Srini:~$ sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
manchodu@Srini:~$ sudo ip addr add 192.168.20.2/24 dev eth0
manchodu@Srini:~$ sudo ip route add 192.168.20.0/24 via 192.168.10.1
RTNETLINK answers: File exists
manchodu@Srini:~$ sudo ip route add 192.168.10.0/24 via 192.168.20.1
RTNETLINK answers: File exists
manchodu@Srini:~$ ping 192.168.20.2
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=64 time=0.034 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=64 time=0.036 ms
64 bytes from 192.168.20.2: icmp_seq=3 ttl=64 time=0.053 ms
64 bytes from 192.168.20.2: icmp_seq=4 ttl=64 time=0.047 ms
64 bytes from 192.168.20.2: icmp_seq=5 ttl=64 time=0.047 ms
64 bytes from 192.168.20.2: icmp_seq=6 ttl=64 time=0.258 ms
64 bytes from 192.168.20.2: icmp_seq=7 ttl=64 time=0.102 ms
64 bytes from 192.168.20.2: icmp_seq=8 ttl=64 time=0.087 ms
64 bytes from 192.168.20.2: icmp_seq=9 ttl=64 time=0.057 ms
64 bytes from 192.168.20.2: icmp_seq=10 ttl=64 time=0.036 ms
64 bytes from 192.168.20.2: icmp_seq=11 ttl=64 time=0.035 ms
64 bytes from 192.168.20.2: icmp_seq=12 ttl=64 time=0.038 ms
64 bytes from 192.168.20.2: icmp_seq=13 ttl=64 time=0.030 ms
64 bytes from 192.168.20.2: icmp_seq=14 ttl=64 time=0.031 ms
64 bytes from 192.168.20.2: icmp_seq=15 ttl=64 time=0.027 ms
64 bytes from 192.168.20.2: icmp_seq=16 ttl=64 time=0.092 ms
64 bytes from 192.168.20.2: icmp_seq=17 ttl=64 time=0.119 ms
64 bytes from 192.168.20.2: icmp_seq=18 ttl=64 time=0.070 ms
64 bytes from 192.168.20.2: icmp_seq=19 ttl=64 time=0.064 ms
^C
--- 192.168.20.2 ping statistics ---
19 packets transmitted, 19 received, 0% packet loss, time 17991ms
rtt min/avg/max/mdev = 0.027/0.066/0.258/0.052 ms
manchodu@Srini:~$ sudo tc qdisc replace dev eth1 root tbf rate 5mbit burst 32kb limit 64kb
Cannot find device "eth1"
manchodu@Srini:~$ sudo tc -s qdisc show dev eth1
Cannot find device "eth1"
manchodu@Srini:~$ watch -n 1 'tc -s qdisc show dev eth1'
manchodu@Srini:~$ iperf3 -s
iperf3: error - unable to start listener for connections: Address already in use
iperf3: exiting
manchodu@Srini:~$ iperf3 -c 192.168.20.2 -u -b 1M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.20.2 port 52517 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   122 KBytes   998 Kbits/sec  105
[  5]   1.00-2.00   sec   123 KBytes  1.01 Mbits/sec  104
[  5]   2.00-3.00   sec   122 KBytes   998 Kbits/sec  104
```

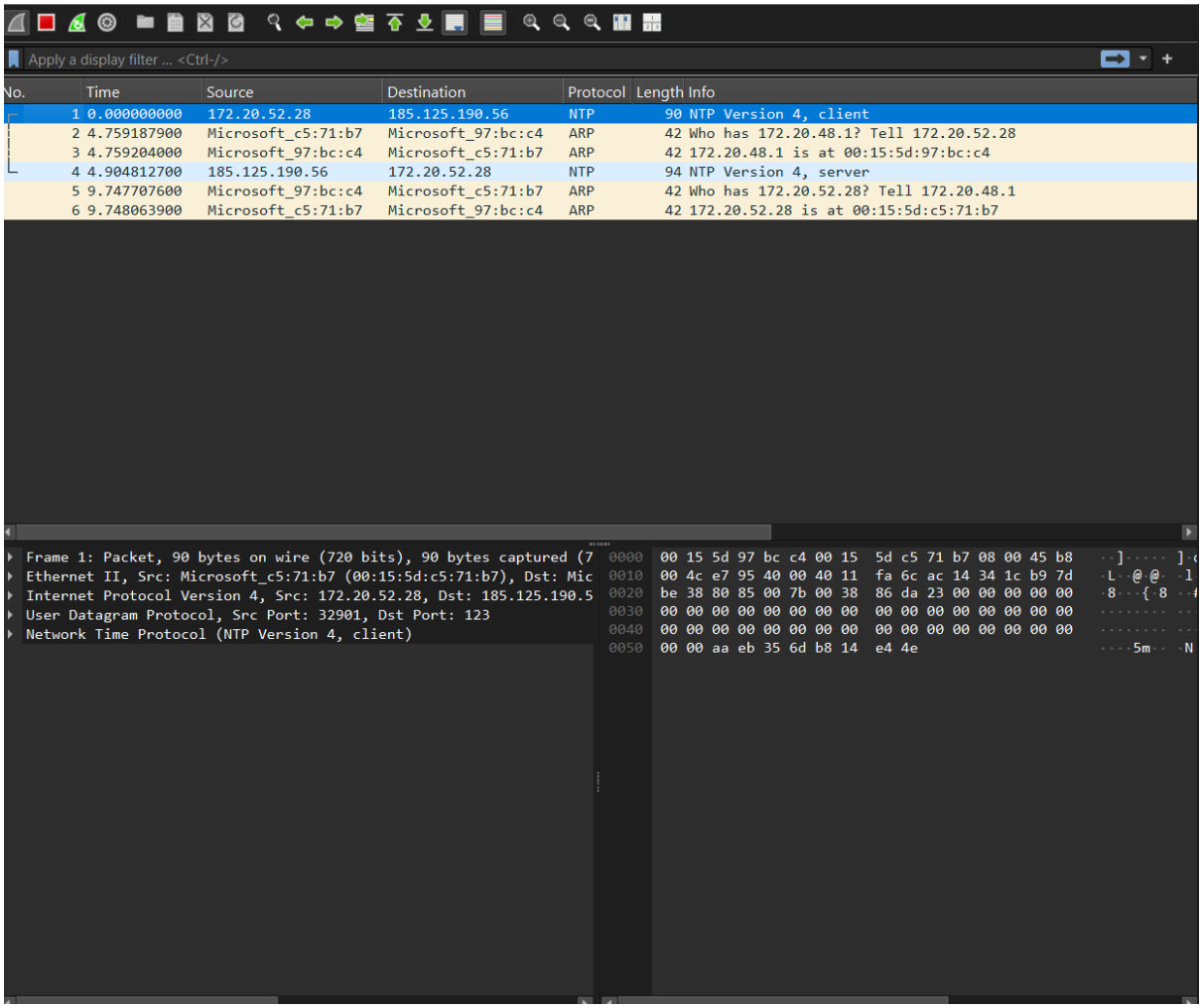
```
manchodu@Srini:~$ iperf3 -c 192.168.20.2 -u -b 2M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.20.2 port 55633 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   245 KBytes  2.01 Mbits/sec  209
[  5]   1.00-2.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   2.00-3.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   3.00-4.00   sec   245 KBytes  2.01 Mbits/sec  209
^C[  5]   4.00-4.99   sec   240 KBytes  1.99 Mbits/sec  205
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-4.99   sec  1.19 MBytes  2.00 Mbits/sec  0.000 ms  0/1039 (0%)  sender
[  5]   0.00-4.99   sec  0.00 Bytes  0.00 bits/sec  0.000 ms  0/0 (0%)  receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
```

```
manchodu@Srini:~$ iperf3 -c 192.168.20.2 -u -b 4M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.20.2 port 50267 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   489 KBytes  4.00 Mbits/sec  417
[  5]   1.00-2.00   sec   488 KBytes  3.99 Mbits/sec  416
[  5]   2.00-3.00   sec   489 KBytes  4.00 Mbits/sec  417
[  5]   3.00-4.00   sec   489 KBytes  4.00 Mbits/sec  417
[  5]   4.00-5.00   sec   488 KBytes  3.99 Mbits/sec  416
[  5]   5.00-6.00   sec   489 KBytes  4.00 Mbits/sec  417
^C[  5]   6.00-6.16   sec  76.2 KBytes  3.97 Mbits/sec  65
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-6.16   sec  2.94 MBytes  4.00 Mbits/sec  0.000 ms  0/2565 (0%)  sender
[  5]   0.00-6.16   sec  0.00 Bytes  0.00 bits/sec  0.000 ms  0/0 (0%)  receiver
iperf3: interrupt - the client has terminated
```

```
manchodu@Srini:~$ iperf3 -c 192.168.20.2 -u -b 5M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.20.2 port 42674 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   1.00-2.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   2.00-3.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   3.00-4.00   sec   609 KBytes  4.99 Mbits/sec  520
[  5]   4.00-5.00   sec   611 KBytes  5.00 Mbits/sec  521
^C[  5]   5.00-5.18   sec   109 KBytes  5.00 Mbits/sec  93
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-5.18   sec  3.09 MBytes  5.00 Mbits/sec  0.000 ms  0/2697 (0%)  sender
[  5]   0.00-5.18   sec  0.00 Bytes  0.00 bits/sec  0.000 ms  0/0 (0%)  receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$ iperf3 -c 192.168.20.2 -u -b 7M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.20.2 port 59562 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   1.00-2.00   sec   854 KBytes  7.00 Mbits/sec  729
^C[  5]   2.00-2.36   sec   309 KBytes  7.02 Mbits/sec  264
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-2.36   sec  1.97 MBytes  7.00 Mbits/sec  0.000 ms  0/1722 (0%)  sender
[  5]   0.00-2.36   sec  0.00 Bytes  0.00 bits/sec  0.000 ms  0/0 (0%)  receiver
iperf3: interrupt - the client has terminated
```

```
manchodu@Srini:~$ iperf3 -c 192.168.20.2 -u -b 10M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.20.2 port 45206 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]   1.00-2.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   2.00-3.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
^C[  5]   3.00-3.59   sec   724 KBytes  9.99 Mbits/sec  618
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-3.59   sec  4.28 MBytes  10.0 Mbits/sec  0.000 ms  0/3743 (0%)  sender
[  5]   0.00-3.59   sec  0.00 Bytes  0.00 bits/sec  0.000 ms  0/0 (0%)  receiver
```


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Welcome to Wireshark
4.6.9 (v4.6.9-0-g2d548b197c75). You receive automatic updates.
Open
C:\Users\thath_8qk\Documents\Wireshark\Lab1_Capture.pcapng..pcapng (116 MB)
CERTIFIED ANALYST
WIRESHARK
WCA
Wireshark Certification
Wireshark Certified Analyst Certification
Designed to formally recognize engineers who demonstrate professional-level proficiency in network protocol analysis and troubleshooting using Wireshark.
More Info
Capture
...using this filter:
Enter a capture filter ...
All interfaces shown
vEthernet (WSL (Hyper-V firewall))
Wi-Fi
Adapter for loopback traffic capture
Local Area Connection* 10
Local Area Connection* 9
Local Area Connection* 8
Local Area Connection* 2
Local Area Connection* 1
USBPcap1: \\.\USBPcap1
USBPcap2: \\.\USBPcap2
Event Tracing for Windows (ETW) reader: etwdump
Learn
User Documentation
Wiki
Questions and Answers
Mailing Lists
Discord
Donate




```
manchodu@Srini:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:c5:71:b7 brd ff:ff:ff:ff:ff:ff
    inet 172.20.52.28/20 brd 172.20.63.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.10.2/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.20.2/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.10.1/24 scope global secondary eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fec5:71b7/64 scope link
```

```
manchodu@Srini:~$ sudo ip addr del 192.168.10.2/24 dev eth0
manchodu@Srini:~$ sudo ip addr del 192.168.20.2/24 dev eth0
manchodu@Srini:~$ sudo ip addr del 192.168.10.1/24 dev eth0
Error: ipv4: Address not found.
```

```
manchodu@Srini:~$ ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:c5:71:b7 brd ff:ff:ff:ff:ff:ff
    inet 172.20.52.28/20 brd 172.20.63.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fec5:71b7/64 scope link
       valid_lft forever preferred_lft forever
```

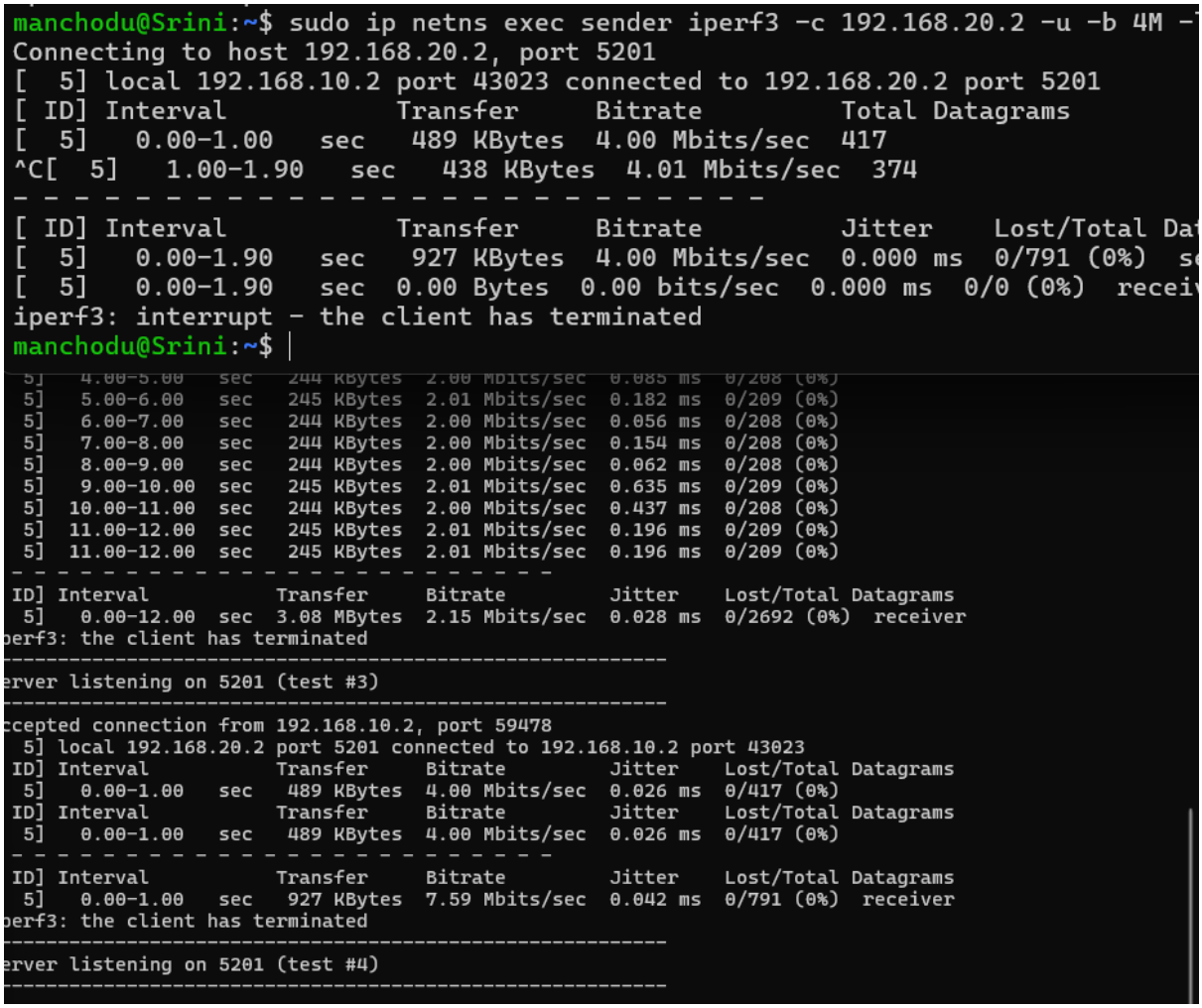
```
manchodu@Srini:~$ sudo ip netns exec router tc -s qdisc show dev r1
qdisc tbf 8001: root refcnt 13 rate 5Mbit burst 32Kb lat 52.4ms
 Sent 0 bytes 0 pkt (dropped 0, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
```

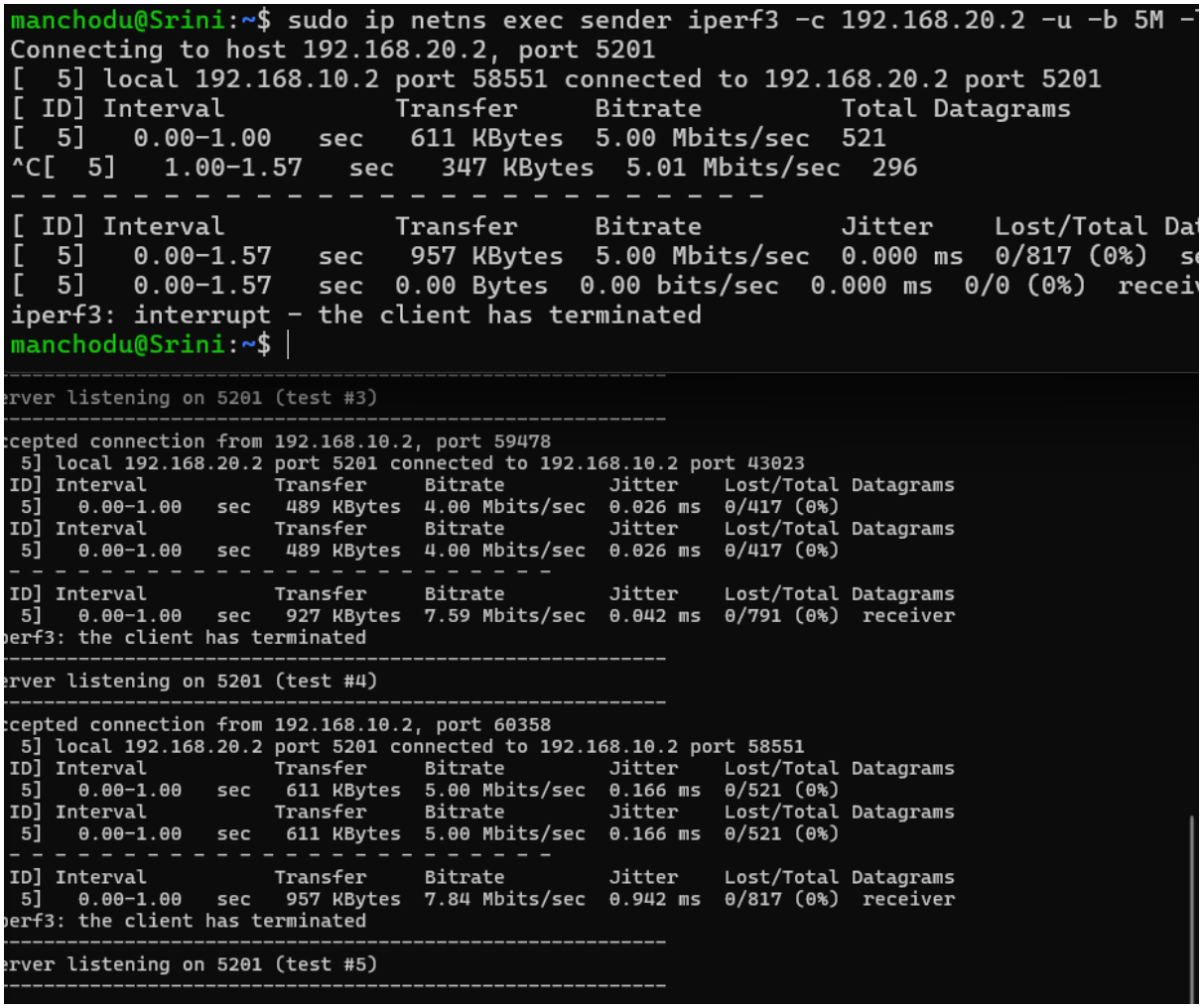
```
manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 1M -l 1200 -t 30 -i 1
iperf3: error - unable to connect to server - server may have stopped running or use a different port,
 firewall issue, etc.: Connection refused
manchodu@Srini:~$
```

```
manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 1M -l 1200
[sudo] password for manchodu:
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 41639 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  105
[  5]   1.00-2.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   2.00-3.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   3.00-4.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   4.00-5.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   5.00-6.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   6.00-7.00   sec   123 KBytes  1.01 Mbits/sec  105
[  5]   7.00-8.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   8.00-9.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   9.00-10.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  10.00-11.00  sec   122 KBytes   998 Kbits/sec  104
```

```
Server listening on 5201 (test #1)
-----------------------------------------------------------
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 1M -l 1200 -t 30 -i 1
Accepted connection from 192.168.10.2, port 38460
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 41639
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  0.241 ms  0/105 (0%)
[  5]   1.00-2.00   sec   122 KBytes   998 Kbits/sec  1.056 ms  0/104 (0%)
[  5]   2.00-3.00   sec   122 KBytes   998 Kbits/sec  0.723 ms  0/104 (0%)
[  5]   3.00-4.00   sec   122 KBytes   998 Kbits/sec  0.984 ms  0/104 (0%)
[  5]   4.00-5.00   sec   122 KBytes   998 Kbits/sec  0.782 ms  0/104 (0%)
[  5]   5.00-6.00   sec   123 KBytes  1.01 Mbits/sec  0.330 ms  0/105 (0%)
[  5]   6.00-7.00   sec   122 KBytes   999 Kbits/sec  0.116 ms  0/104 (0%)
[  5]   7.00-8.00   sec   122 KBytes   998 Kbits/sec  0.069 ms  0/104 (0%)
[  5]   8.00-9.00   sec   122 KBytes   998 Kbits/sec  0.339 ms  0/104 (0%)
[  5]   9.00-10.00  sec   122 KBytes   998 Kbits/sec  0.067 ms  0/104 (0%)
[  5]  10.00-11.00  sec   122 KBytes   998 Kbits/sec  0.112 ms  0/104 (0%)
```

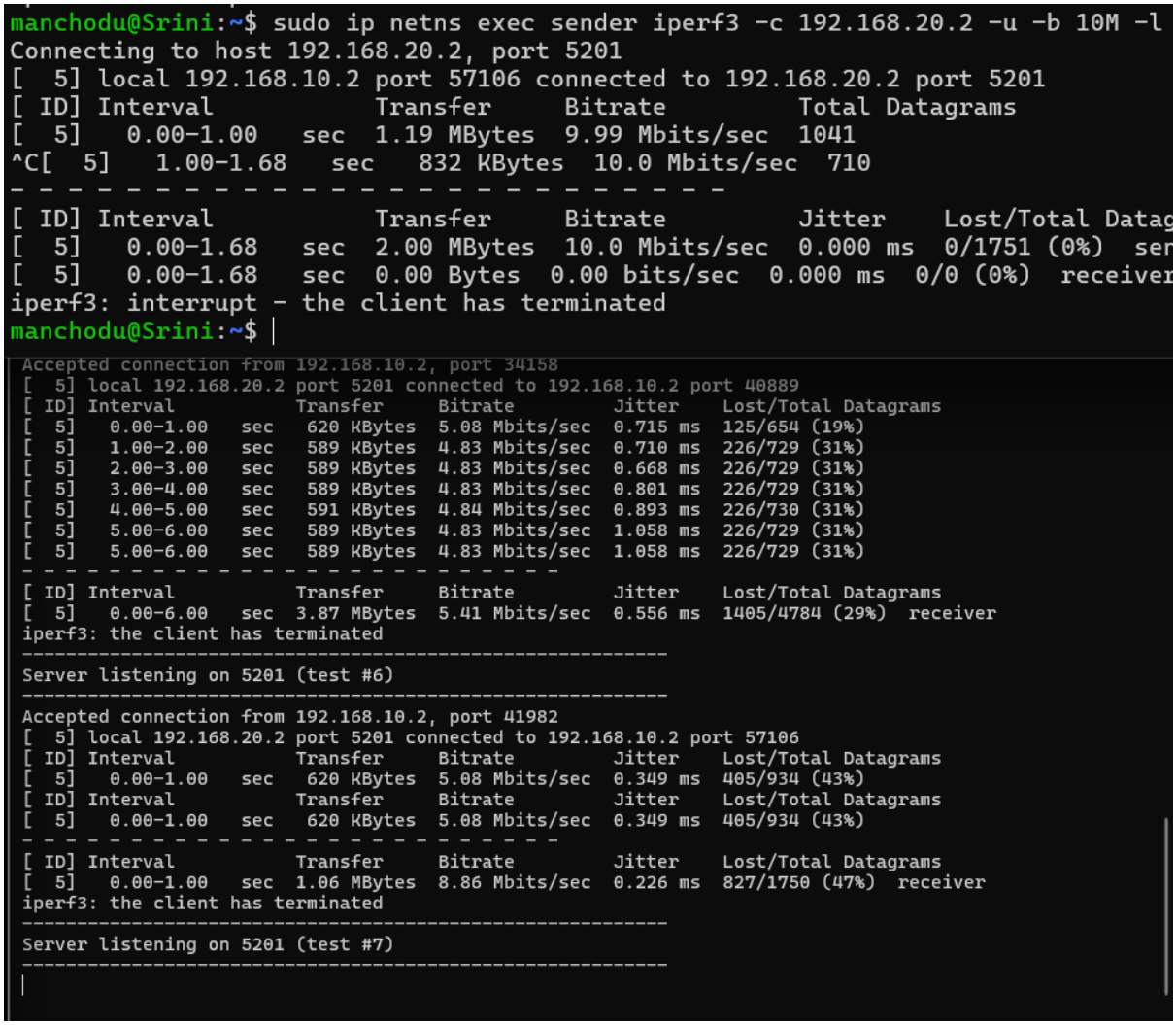
```
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 40911 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   245 KBytes  2.01 Mbits/sec  209
[  5]   1.00-2.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   2.00-3.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   3.00-4.00   sec   245 KBytes  2.01 Mbits/sec  209
[  5]   4.00-5.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   5.00-6.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   6.00-7.00   sec   245 KBytes  2.01 Mbits/sec  209
[  5]   7.00-8.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   8.00-9.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   9.00-10.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  10.00-11.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  11.00-12.00  sec   244 KBytes  2.00 Mbits/sec  208
^C[  5]  12.00-12.92  sec   226 KBytes  2.01 Mbits/sec  193
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Da
[  5]   0.00-12.92  sec  3.08 MBytes  2.00 Mbits/sec  0.000 ms  0/2693 (0%)
[  5]   0.00-12.92  sec  0.00 Bytes  0.00 bits/sec  0.000 ms  0/0 (0%)  recei
iperf3: interrupt - the client has terminated
 ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
  5]   0.00-19.00  sec  2.35 MBytes  1.04 Mbits/sec  1.642 ms  0/2054 (0%)  receiver
perf3: the client has terminated
-----------------------------------------------------------
erver listening on 5201 (test #2)
-----------------------------------------------------------
ccepted connection from 192.168.10.2, port 33748
  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 40911
 ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
  5]   0.00-1.00   sec   244 KBytes  2.00 Mbits/sec  2.134 ms  0/208 (0%)
  5]   1.00-2.00   sec   245 KBytes  2.01 Mbits/sec  0.456 ms  0/209 (0%)
  5]   2.00-3.00   sec   244 KBytes  1.99 Mbits/sec  1.553 ms  0/208 (0%)
  5]   3.00-4.00   sec   245 KBytes  2.01 Mbits/sec  0.055 ms  0/209 (0%)
  5]   4.00-5.00   sec   244 KBytes  2.00 Mbits/sec  0.085 ms  0/208 (0%)
  5]   5.00-6.00   sec   245 KBytes  2.01 Mbits/sec  0.182 ms  0/209 (0%)
  5]   6.00-7.00   sec   244 KBytes  2.00 Mbits/sec  0.056 ms  0/208 (0%)
  5]   7.00-8.00   sec   244 KBytes  2.00 Mbits/sec  0.154 ms  0/208 (0%)
  5]   8.00-9.00   sec   244 KBytes  2.00 Mbits/sec  0.062 ms  0/208 (0%)
  5]   9.00-10.00  sec   245 KBytes  2.01 Mbits/sec  0.635 ms  0/209 (0%)
  5]  10.00-11.00  sec   244 KBytes  2.00 Mbits/sec  0.437 ms  0/208 (0%)
  5]  11.00-12.00  sec   245 KBytes  2.01 Mbits/sec  0.196 ms  0/209 (0%)
  5]  11.00-12.00  sec   245 KBytes  2.01 Mbits/sec  0.196 ms  0/209 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
 ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
  5]   0.00-12.00  sec  3.08 MBytes  2.15 Mbits/sec  0.028 ms  0/2692 (0%)  receiver
perf3: the client has terminated
-----------------------------------------------------------
erver listening on 5201 (test #3)
-----------------------------------------------------------
```

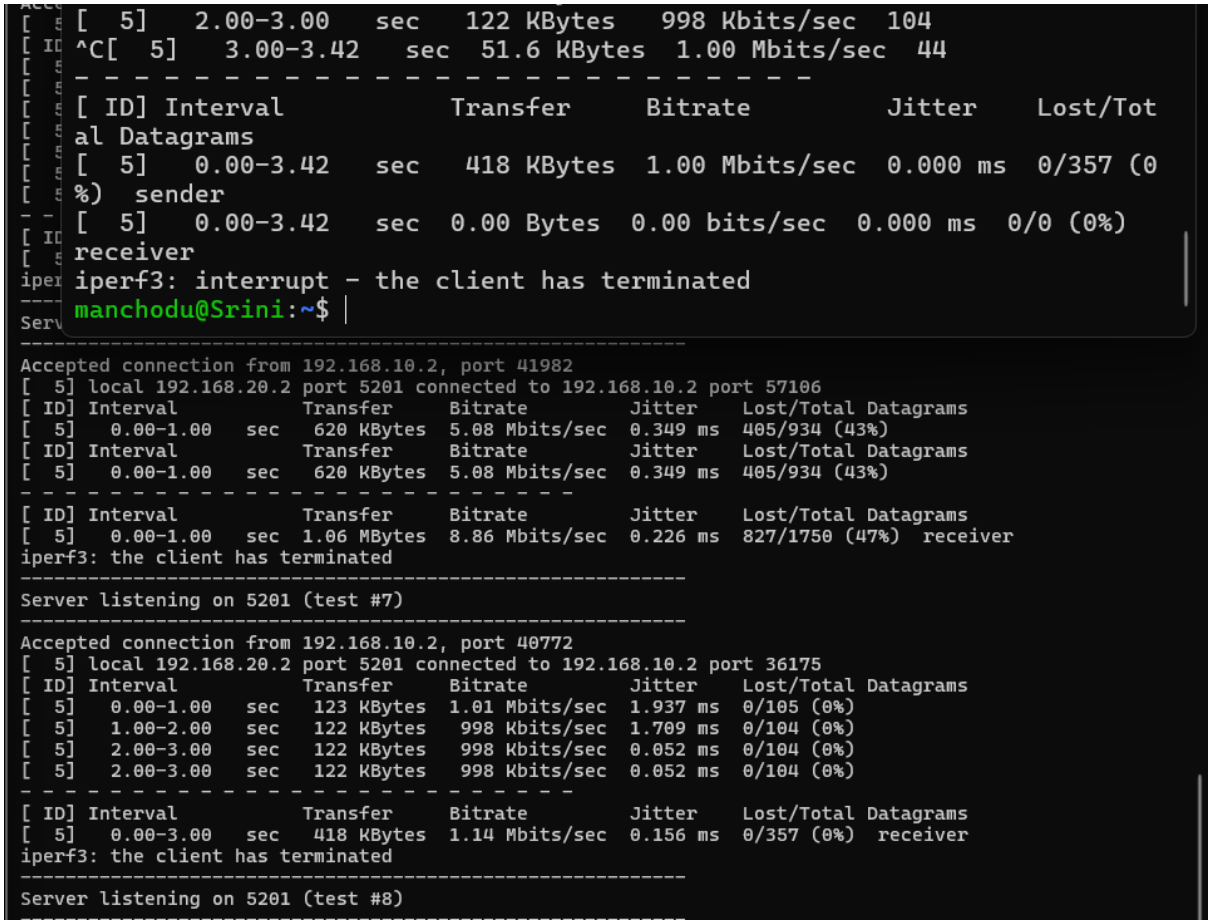

manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 4M -
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 43023 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 489 KBytes 4.00 Mbits/sec 417
^C[5] 1.00-1.90 sec 438 KBytes 4.01 Mbits/sec 374
[ID] Interval Transfer Bitrate Jitter Lost/Total Dat
[5] 0.00-1.90 sec 927 KBytes 4.00 Mbits/sec 0.000 ms 0/791 (0%) s
[5] 0.00-1.90 sec 0.00 Bytes 0.00 bits/sec 0.000 ms 0/0 (0%) recei
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
5] 4.00-5.00 sec 244 KBytes 2.00 Mbits/sec 0.085 ms 0/208 (0%)
5] 5.00-6.00 sec 245 KBytes 2.01 Mbits/sec 0.182 ms 0/209 (0%)
5] 6.00-7.00 sec 244 KBytes 2.00 Mbits/sec 0.056 ms 0/208 (0%)
5] 7.00-8.00 sec 244 KBytes 2.00 Mbits/sec 0.154 ms 0/208 (0%)
5] 8.00-9.00 sec 244 KBytes 2.00 Mbits/sec 0.062 ms 0/208 (0%)
5] 9.00-10.00 sec 245 KBytes 2.01 Mbits/sec 0.635 ms 0/209 (0%)
5] 10.00-11.00 sec 244 KBytes 2.00 Mbits/sec 0.437 ms 0/208 (0%)
5] 11.00-12.00 sec 245 KBytes 2.01 Mbits/sec 0.196 ms 0/209 (0%)
5] 11.00-12.00 sec 245 KBytes 2.01 Mbits/sec 0.196 ms 0/209 (0%)
ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
5] 0.00-12.00 sec 3.08 MBytes 2.15 Mbits/sec 0.028 ms 0/2692 (0%) receiver
erf3: the client has terminated
erver listening on 5201 (test #3)
ccepted connection from 192.168.10.2, port 59478
5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 43023
ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
5] 0.00-1.00 sec 489 KBytes 4.00 Mbits/sec 0.026 ms 0/417 (0%)
ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
5] 0.00-1.00 sec 489 KBytes 4.00 Mbits/sec 0.026 ms 0/417 (0%)
ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
5] 0.00-1.00 sec 927 KBytes 7.59 Mbits/sec 0.042 ms 0/791 (0%) receiver
erf3: the client has terminated
erver listening on 5201 (test #4)


manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 5M -
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 58551 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 611 KBytes 5.00 Mbits/sec 521
^C[5] 1.00-1.57 sec 347 KBytes 5.01 Mbits/sec 296
[ID] Interval Transfer Bitrate Jitter Lost/Total Da
[5] 0.00-1.57 sec 957 KBytes 5.00 Mbits/sec 0.000 ms 0/817 (0%) s
[5] 0.00-1.57 sec 0.00 Bytes 0.00 bits/sec 0.000 ms 0/0 (0%) recei
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
rver listening on 5201 (test #3)
cepted connection from 192.168.10.2, port 59478
5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 43023
ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
5] 0.00-1.00 sec 489 KBytes 4.00 Mbits/sec 0.026 ms 0/417 (0%)
ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
5] 0.00-1.00 sec 489 KBytes 4.00 Mbits/sec 0.026 ms 0/417 (0%)
ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
5] 0.00-1.00 sec 927 KBytes 7.59 Mbits/sec 0.042 ms 0/791 (0%) receiver
erf3: the client has terminated
rver listening on 5201 (test #4)
cepted connection from 192.168.10.2, port 60358
5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 58551
ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
5] 0.00-1.00 sec 611 KBytes 5.00 Mbits/sec 0.166 ms 0/521 (0%)
ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
5] 0.00-1.00 sec 611 KBytes 5.00 Mbits/sec 0.166 ms 0/521 (0%)
ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
5] 0.00-1.00 sec 957 KBytes 7.84 Mbits/sec 0.942 ms 0/817 (0%) receiver
erf3: the client has terminated
rver listening on 5201 (test #5)

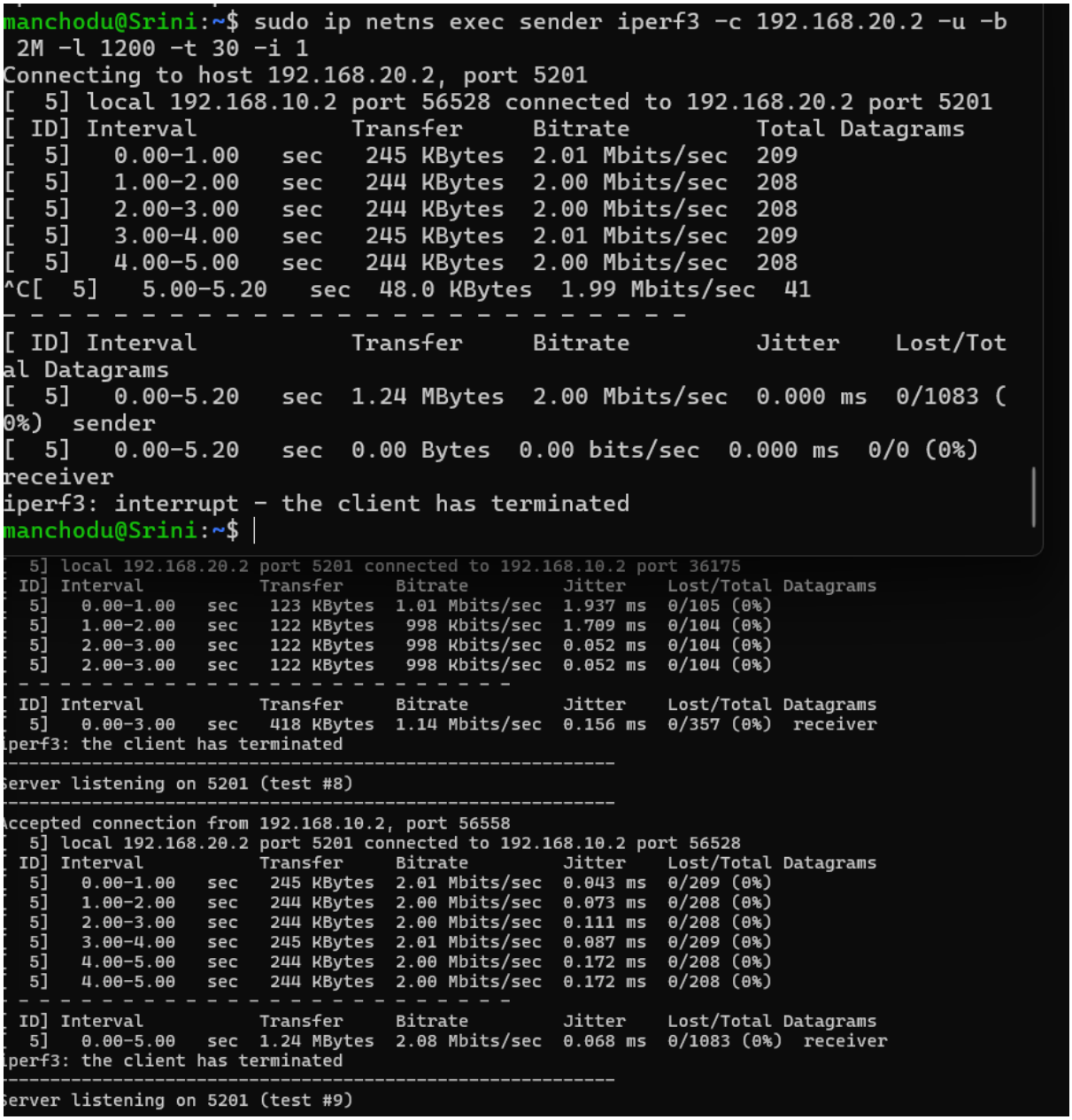
```
manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 7M -l 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 40889 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   1.00-2.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   2.00-3.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   3.00-4.00   sec   855 KBytes  7.01 Mbits/sec  730
^X[  5]   4.00-5.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   5.00-6.00   sec   854 KBytes  7.00 Mbits/sec  729
^C[  5]   6.00-6.56   sec   480 KBytes  7.00 Mbits/sec  410
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datag
[  5]   0.00-6.56   sec  5.48 MBytes  7.00 Mbits/sec  0.000 ms  0/4785 (0%)  ser
[  5]   0.00-6.56   sec  0.00 Bytes  0.00 bits/sec  0.000 ms  0/0 (0%)  receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
```

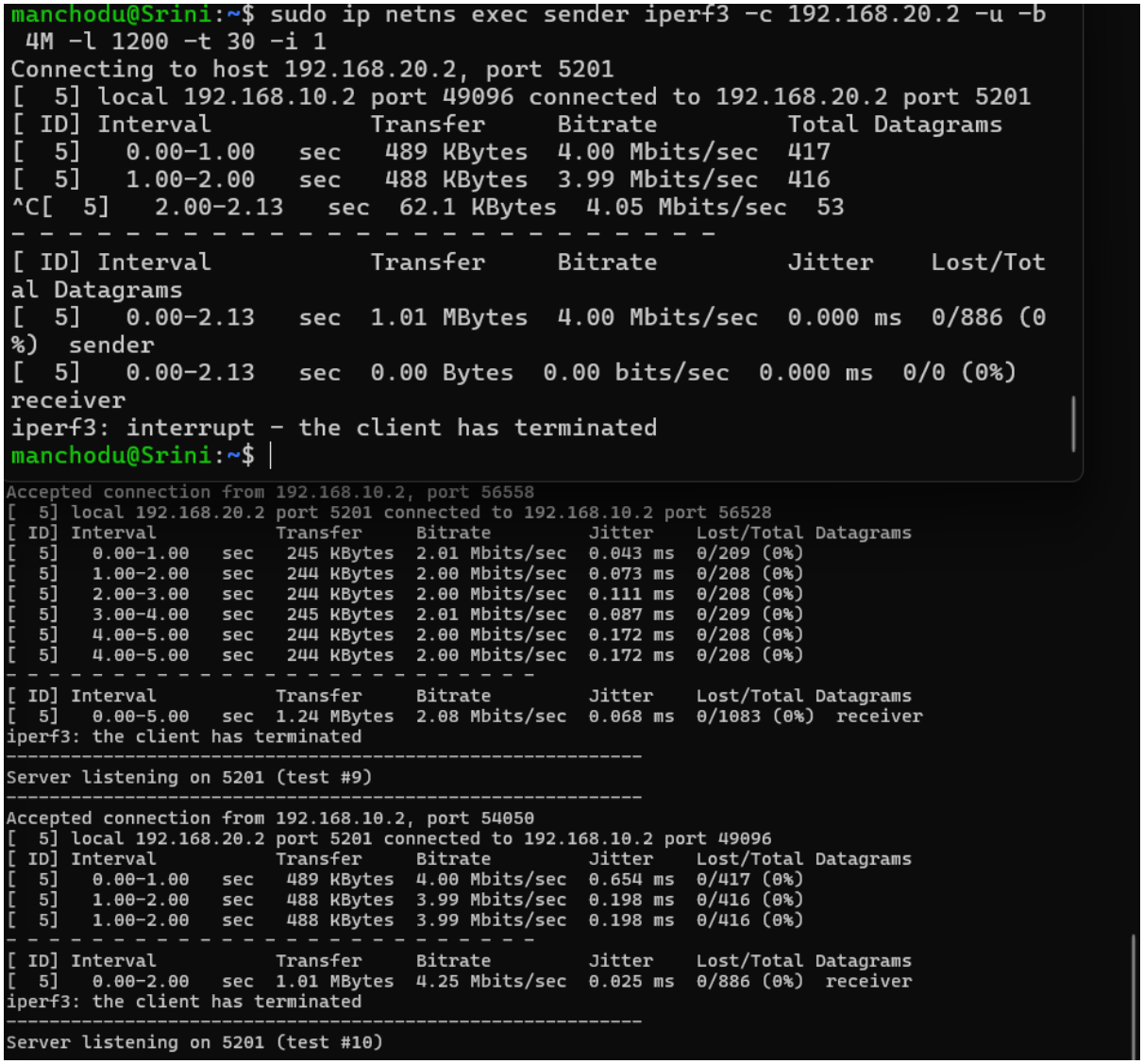
```
Accepted connection from 192.168.10.2, port 60358
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 58551
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   611 KBytes  5.00 Mbits/sec  0.166 ms  0/521 (0%)
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   611 KBytes  5.00 Mbits/sec  0.166 ms  0/521 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   957 KBytes  7.84 Mbits/sec  0.942 ms  0/817 (0%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #5)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 34158
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 40889
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   620 KBytes  5.08 Mbits/sec  0.715 ms  125/654 (19%)
[  5]   1.00-2.00   sec   589 KBytes  4.83 Mbits/sec  0.710 ms  226/729 (31%)
[  5]   2.00-3.00   sec   589 KBytes  4.83 Mbits/sec  0.668 ms  226/729 (31%)
[  5]   3.00-4.00   sec   589 KBytes  4.83 Mbits/sec  0.801 ms  226/729 (31%)
[  5]   4.00-5.00   sec   591 KBytes  4.84 Mbits/sec  0.893 ms  226/730 (31%)
[  5]   5.00-6.00   sec   589 KBytes  4.83 Mbits/sec  1.058 ms  226/729 (31%)
[  5]   5.00-6.00   sec   589 KBytes  4.83 Mbits/sec  1.058 ms  226/729 (31%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-6.00   sec  3.87 MBytes  5.41 Mbits/sec  0.556 ms  1405/4784 (29%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #6)
-----------------------------------------------------------
```

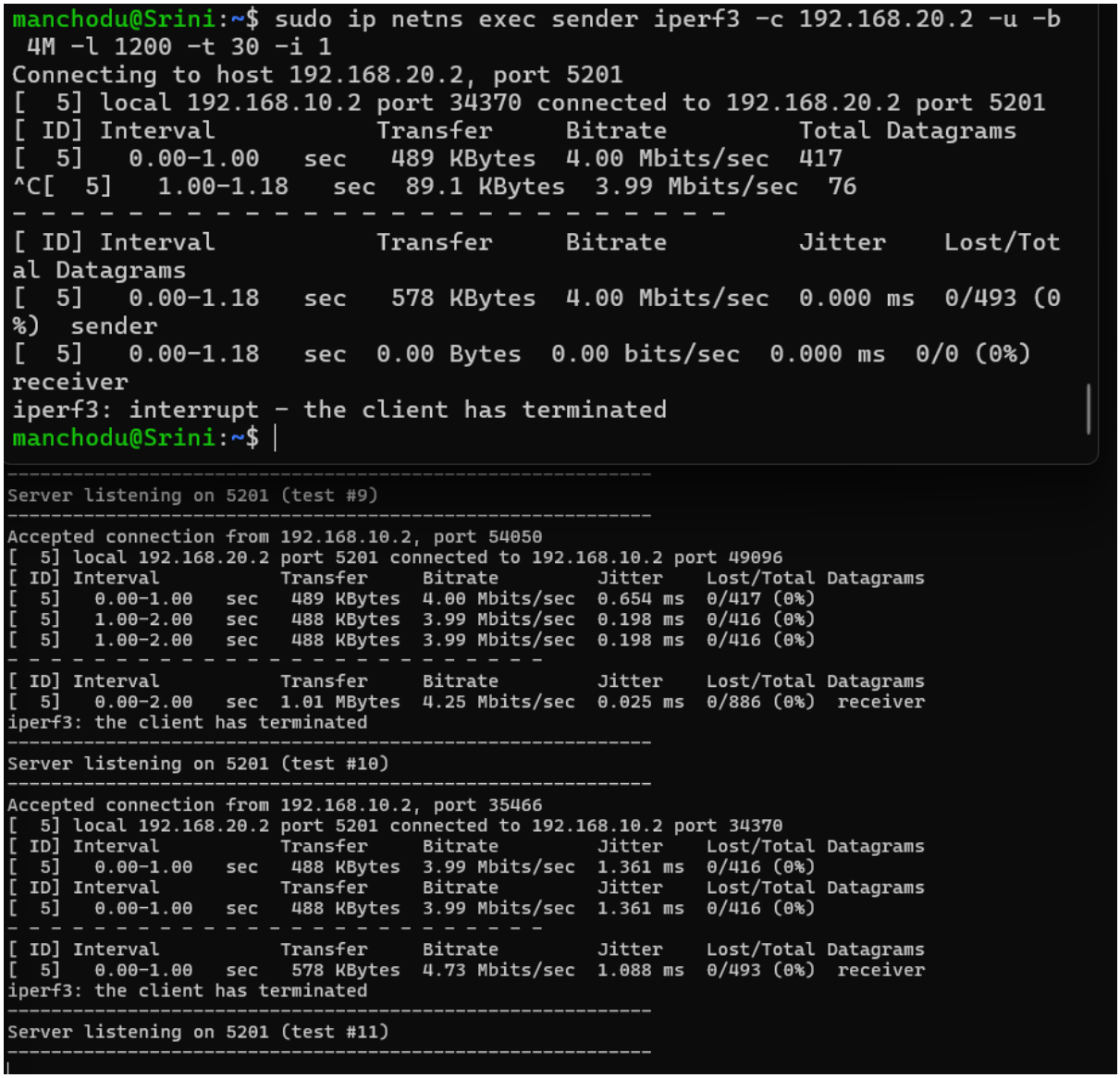

manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 10M -l
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 57106 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 1.19 MBytes 9.99 Mbits/sec 1041
^C[5] 1.00-1.68 sec 832 KBytes 10.0 Mbits/sec 710
[ID] Interval Transfer Bitrate Jitter Lost/Total Datag
[5] 0.00-1.68 sec 2.00 MBytes 10.0 Mbits/sec 0.000 ms 0/1751 (0%) sen
[5] 0.00-1.68 sec 0.00 Bytes 0.00 bits/sec 0.000 ms 0/0 (0%) receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
Accepted connection from 192.168.10.2, port 34158
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 40889
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 620 KBytes 5.08 Mbits/sec 0.715 ms 125/654 (19%)
[5] 1.00-2.00 sec 589 KBytes 4.83 Mbits/sec 0.710 ms 226/729 (31%)
[5] 2.00-3.00 sec 589 KBytes 4.83 Mbits/sec 0.668 ms 226/729 (31%)
[5] 3.00-4.00 sec 589 KBytes 4.83 Mbits/sec 0.801 ms 226/729 (31%)
[5] 4.00-5.00 sec 591 KBytes 4.84 Mbits/sec 0.893 ms 226/730 (31%)
[5] 5.00-6.00 sec 589 KBytes 4.83 Mbits/sec 1.058 ms 226/729 (31%)
[5] 5.00-6.00 sec 589 KBytes 4.83 Mbits/sec 1.058 ms 226/729 (31%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-6.00 sec 3.87 MBytes 5.41 Mbits/sec 0.556 ms 1405/4784 (29%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #6)
Accepted connection from 192.168.10.2, port 41982
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 57106
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 620 KBytes 5.08 Mbits/sec 0.349 ms 405/934 (43%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 620 KBytes 5.08 Mbits/sec 0.349 ms 405/934 (43%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 1.06 MBytes 8.86 Mbits/sec 0.226 ms 827/1750 (47%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #7)



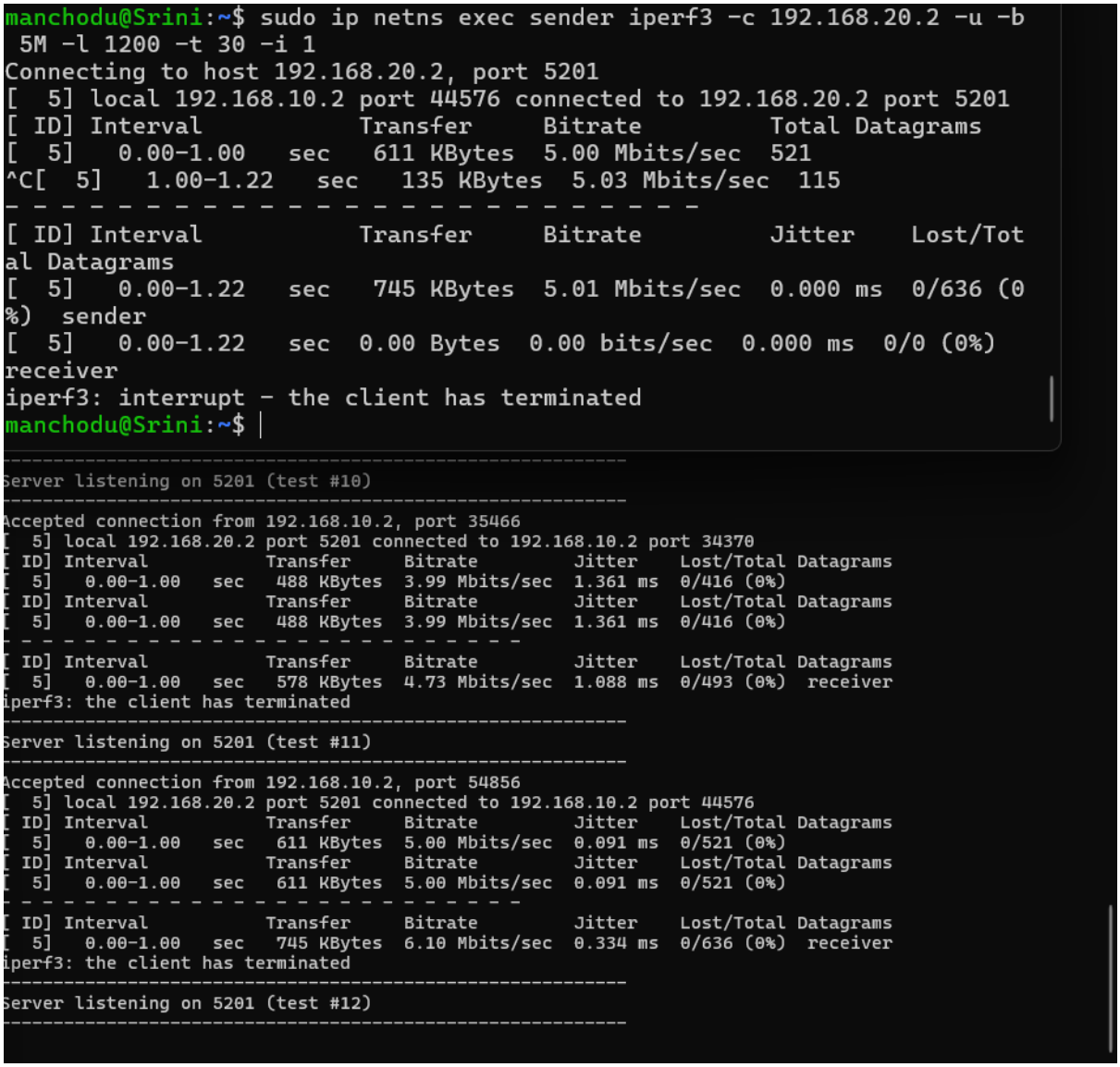

```
Accepted connection from 192.168.10.2, port 41982
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 57106
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   620 KBytes  5.08 Mbits/sec  0.349 ms  405/934 (43%)
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   620 KBytes  5.08 Mbits/sec  0.349 ms  405/934 (43%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec  1.06 MBytes  8.86 Mbits/sec  0.226 ms  827/1750 (47%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #7)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 40772
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 36175
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  1.937 ms  0/105 (0%)
[  5]   1.00-2.00   sec   122 KBytes   998 Kbits/sec  1.709 ms  0/104 (0%)
[  5]   2.00-3.00   sec   122 KBytes   998 Kbits/sec  0.052 ms  0/104 (0%)
[  5]   2.00-3.00   sec   122 KBytes   998 Kbits/sec  0.052 ms  0/104 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-3.00   sec   418 KBytes  1.14 Mbits/sec  0.156 ms  0/357 (0%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #8)
-----------------------------------------------------------
```

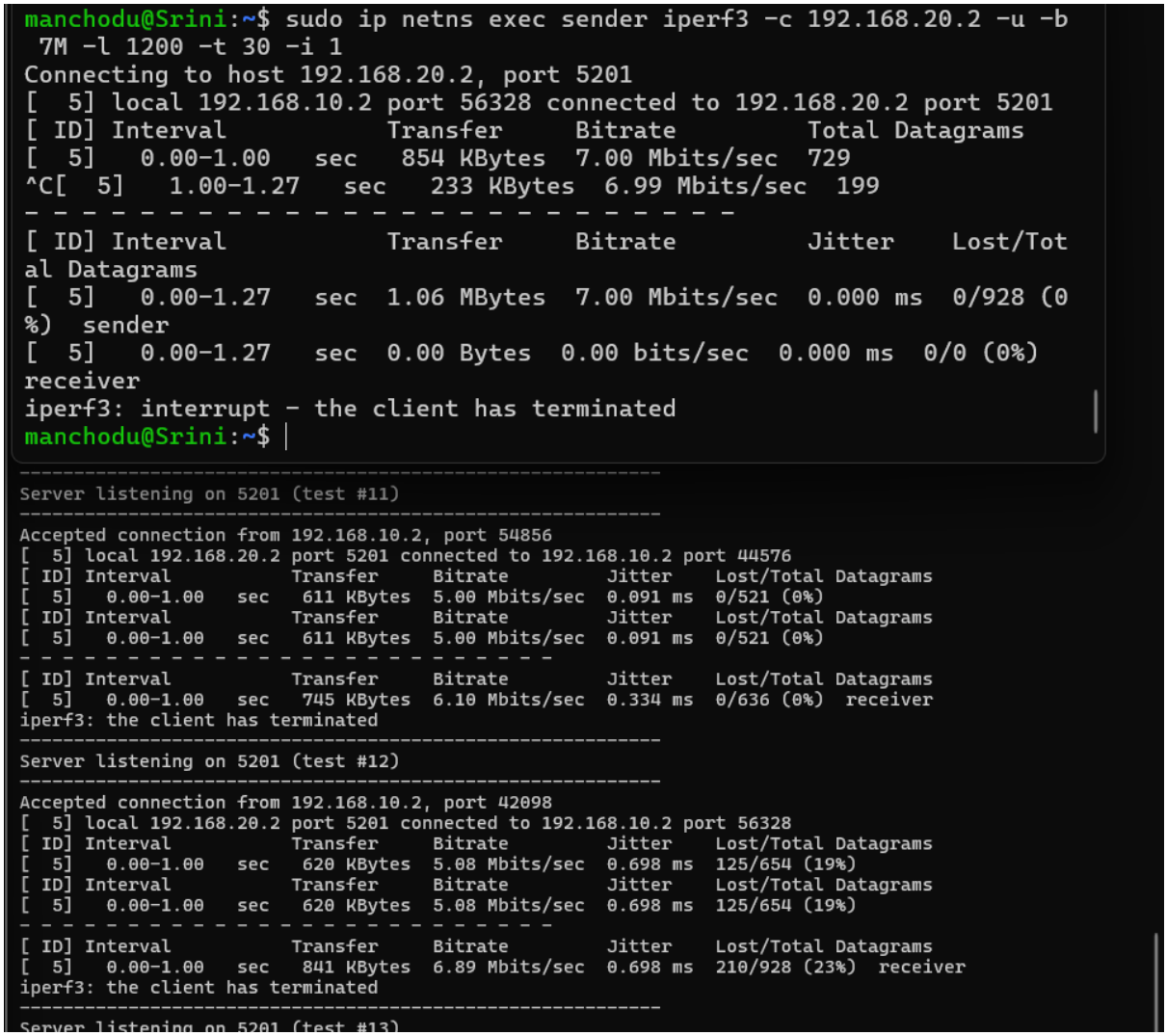

manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 2M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 56528 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 1.00-2.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 2.00-3.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 3.00-4.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 4.00-5.00 sec 244 KBytes 2.00 Mbits/sec 208
^C[5] 5.00-5.20 sec 48.0 KBytes 1.99 Mbits/sec 41
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-5.20 sec 1.24 MBytes 2.00 Mbits/sec 0.000 ms 0/1083 (0%) sender
[5] 0.00-5.20 sec 0.00 Bytes 0.00 bits/sec 0.000 ms 0/0 (0%) receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 36175
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 123 KBytes 1.01 Mbits/sec 1.937 ms 0/105 (0%)
[5] 1.00-2.00 sec 122 KBytes 998 Kbits/sec 1.709 ms 0/104 (0%)
[5] 2.00-3.00 sec 122 KBytes 998 Kbits/sec 0.052 ms 0/104 (0%)
[5] 2.00-3.00 sec 122 KBytes 998 Kbits/sec 0.052 ms 0/104 (0%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-3.00 sec 418 KBytes 1.14 Mbits/sec 0.156 ms 0/357 (0%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #8)
Accepted connection from 192.168.10.2, port 56558
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 56528
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 245 KBytes 2.01 Mbits/sec 0.043 ms 0/209 (0%)
[5] 1.00-2.00 sec 244 KBytes 2.00 Mbits/sec 0.073 ms 0/208 (0%)
[5] 2.00-3.00 sec 244 KBytes 2.00 Mbits/sec 0.111 ms 0/208 (0%)
[5] 3.00-4.00 sec 245 KBytes 2.01 Mbits/sec 0.087 ms 0/209 (0%)
[5] 4.00-5.00 sec 244 KBytes 2.00 Mbits/sec 0.172 ms 0/208 (0%)
[5] 4.00-5.00 sec 244 KBytes 2.00 Mbits/sec 0.172 ms 0/208 (0%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-5.00 sec 1.24 MBytes 2.08 Mbits/sec 0.068 ms 0/1083 (0%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #9)


manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 4M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 49096 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 1.00-2.00 sec 488 KBytes 3.99 Mbits/sec 416
^C[5] 2.00-2.13 sec 62.1 KBytes 4.05 Mbits/sec 53
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-2.13 sec 1.01 MBytes 4.00 Mbits/sec 0.000 ms 0/886 (0%) sender
[5] 0.00-2.13 sec 0.00 Bytes 0.00 bits/sec 0.000 ms 0/0 (0%) receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
Accepted connection from 192.168.10.2, port 56558
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 56528
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 245 KBytes 2.01 Mbits/sec 0.043 ms 0/209 (0%)
[5] 1.00-2.00 sec 244 KBytes 2.00 Mbits/sec 0.073 ms 0/208 (0%)
[5] 2.00-3.00 sec 244 KBytes 2.00 Mbits/sec 0.111 ms 0/208 (0%)
[5] 3.00-4.00 sec 245 KBytes 2.01 Mbits/sec 0.087 ms 0/209 (0%)
[5] 4.00-5.00 sec 244 KBytes 2.00 Mbits/sec 0.172 ms 0/208 (0%)
[5] 4.00-5.00 sec 244 KBytes 2.00 Mbits/sec 0.172 ms 0/208 (0%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-5.00 sec 1.24 MBytes 2.08 Mbits/sec 0.068 ms 0/1083 (0%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #9)
Accepted connection from 192.168.10.2, port 54050
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 49096
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 489 KBytes 4.00 Mbits/sec 0.654 ms 0/417 (0%)
[5] 1.00-2.00 sec 488 KBytes 3.99 Mbits/sec 0.198 ms 0/416 (0%)
[5] 1.00-2.00 sec 488 KBytes 3.99 Mbits/sec 0.198 ms 0/416 (0%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-2.00 sec 1.01 MBytes 4.25 Mbits/sec 0.025 ms 0/886 (0%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #10)



```
manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b
 4M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 34370 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   489 KBytes  4.00 Mbits/sec  417
^C[  5]   1.00-1.18   sec  89.1 KBytes  3.99 Mbits/sec  76
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Tot
al Datagrams
[  5]   0.00-1.18   sec   578 KBytes  4.00 Mbits/sec  0.000 ms  0/493 (0
%)  sender
[  5]   0.00-1.18   sec  0.00 Bytes  0.00 bits/sec  0.000 ms  0/0 (0%)
receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
-----------------------------------------------------------
Server listening on 5201 (test #9)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 54050
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 49096
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   489 KBytes  4.00 Mbits/sec  0.654 ms  0/417 (0%)
[  5]   1.00-2.00   sec   488 KBytes  3.99 Mbits/sec  0.198 ms  0/416 (0%)
[  5]   1.00-2.00   sec   488 KBytes  3.99 Mbits/sec  0.198 ms  0/416 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-2.00   sec  1.01 MBytes  4.25 Mbits/sec  0.025 ms  0/886 (0%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #10)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 35466
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 34370
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   488 KBytes  3.99 Mbits/sec  1.361 ms  0/416 (0%)
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   488 KBytes  3.99 Mbits/sec  1.361 ms  0/416 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   578 KBytes  4.73 Mbits/sec  1.088 ms  0/493 (0%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #11)
-----------------------------------------------------------
```


manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 5M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 44576 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 611 KBytes 5.00 Mbits/sec 521
^C[5] 1.00-1.22 sec 135 KBytes 5.03 Mbits/sec 115
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.22 sec 745 KBytes 5.01 Mbits/sec 0.000 ms 0/636 (0%) sender
[5] 0.00-1.22 sec 0.00 Bytes 0.00 bits/sec 0.000 ms 0/0 (0%) receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
Server listening on 5201 (test #10)
Accepted connection from 192.168.10.2, port 35466
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 34370
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 488 KBytes 3.99 Mbits/sec 1.361 ms 0/416 (0%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 488 KBytes 3.99 Mbits/sec 1.361 ms 0/416 (0%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 578 KBytes 4.73 Mbits/sec 1.088 ms 0/493 (0%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #11)
Accepted connection from 192.168.10.2, port 54856
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 44576
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 611 KBytes 5.00 Mbits/sec 0.091 ms 0/521 (0%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 611 KBytes 5.00 Mbits/sec 0.091 ms 0/521 (0%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 745 KBytes 6.10 Mbits/sec 0.334 ms 0/636 (0%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #12)


manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 7M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 56328 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 854 KBytes 7.00 Mbits/sec 729
^C[5] 1.00-1.27 sec 233 KBytes 6.99 Mbits/sec 199
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.27 sec 1.06 MBytes 7.00 Mbits/sec 0.000 ms 0/928 (0%) sender
[5] 0.00-1.27 sec 0.00 Bytes 0.00 bits/sec 0.000 ms 0/0 (0%) receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
Server listening on 5201 (test #11)
Accepted connection from 192.168.10.2, port 54856
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 44576
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 611 KBytes 5.00 Mbits/sec 0.091 ms 0/521 (0%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 611 KBytes 5.00 Mbits/sec 0.091 ms 0/521 (0%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 745 KBytes 6.10 Mbits/sec 0.334 ms 0/636 (0%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #12)
Accepted connection from 192.168.10.2, port 42098
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 56328
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 620 KBytes 5.08 Mbits/sec 0.698 ms 125/654 (19%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 620 KBytes 5.08 Mbits/sec 0.698 ms 125/654 (19%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 841 KBytes 6.89 Mbits/sec 0.698 ms 210/928 (23%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #13)

```
manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b
 10M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 37955 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  9.99 Mbits/sec  1041
^C[  5]   1.00-1.97   sec  1.16 MBytes  10.0 Mbits/sec  1016
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Tot
al Datagrams
[  5]   0.00-1.97   sec  2.35 MBytes  10.0 Mbits/sec  0.000 ms  0/2057 (
0%)  sender
[  5]   0.00-1.97   sec  0.00 Bytes  0.00 bits/sec  0.000 ms  0/0 (0%)
receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
```

```
-----------------------------------------------------------
Server listening on 5201 (test #12)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 42098
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 56328
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   620 KBytes  5.08 Mbits/sec  0.698 ms  125/654 (19%)
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   620 KBytes  5.08 Mbits/sec  0.698 ms  125/654 (19%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   841 KBytes  6.89 Mbits/sec  0.698 ms  210/928 (23%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #13)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 52268
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 37955
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   621 KBytes  5.08 Mbits/sec  0.588 ms  406/936 (43%)
[  5]   1.00-2.00   sec   588 KBytes  4.82 Mbits/sec  0.566 ms  538/1040 (52%)
[  5]   1.00-2.00   sec   588 KBytes  4.82 Mbits/sec  0.566 ms  538/1040 (52%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-2.00   sec  1.23 MBytes  5.14 Mbits/sec  0.190 ms  985/2056 (48%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #14)
-----------------------------------------------------------
```

No.	Time	Source	Destination	Protocol	Length	Info
56	0.376011	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [40] Time sent=1752.8497530 length=1200
57	0.385026	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [41] Time sent=1752.8587710 length=1200
58	0.395000	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [42] Time sent=1752.8687440 length=1200
59	0.404916	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [43] Time sent=1752.8786610 length=1200
60	0.413932	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [44] Time sent=1752.8876730 length=1200
61	0.423939	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [45] Time sent=1752.8976810 length=1200
62	0.433088	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [46] Time sent=1752.9068050 length=1200
63	0.442902	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [47] Time sent=1752.9166430 length=1200
64	0.452908	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [48] Time sent=1752.9266440 length=1200
65	0.462028	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [49] Time sent=1752.9357660 length=1200
66	0.472092	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [50] Time sent=1752.9458000 length=1200
67	0.481066	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [51] Time sent=1752.9547890 length=1200
68	0.490953	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [52] Time sent=1752.9646810 length=1200
69	0.501036	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [53] Time sent=1752.9747550 length=1200
70	0.509909	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [54] Time sent=1752.9836500 length=1200
71	0.520029	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [55] Time sent=1752.9937740 length=1200
72	0.528925	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [56] Time sent=1753.0026690 length=1200
73	0.538894	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [57] Time sent=1753.0126230 length=1200

```
▸ Frame 1: Packet, 74 bytes on wire (592 bits), 74 bytes captured (5
▸ Ethernet II, Src: ea:38:0b:63:e9:18 (ea:38:0b:63:e9:18), Dst: 12:c
▸ Internet Protocol Version 4, Src: 192.168.10.2, Dst: 192.168.20.2
▸ Transmission Control Protocol, Src Port: 40772, Dst Port: 5201, Se
```

```
0000  12 c8 36 fc 27 ca ea 38  0b 63 e9 18 08 00 45 00   ··6·'··8 ·c····E·
0010  00 3c da 42 40 00 3f 06  c2 24 c0 a8 0a 02 c0 a8   ·<·B@·?· ·$····
0020  14 02 9f 44 14 51 51 bf  cb 86 00 00 00 00 a0 02   ···D·QQ· ········
0030  fa f0 9f 83 00 00 02 04  05 b4 04 02 08 0a ae 37   ········ ·······7
0040  b5 49 00 00 00 00 01 03  03 0a                    ·I····· ··
```

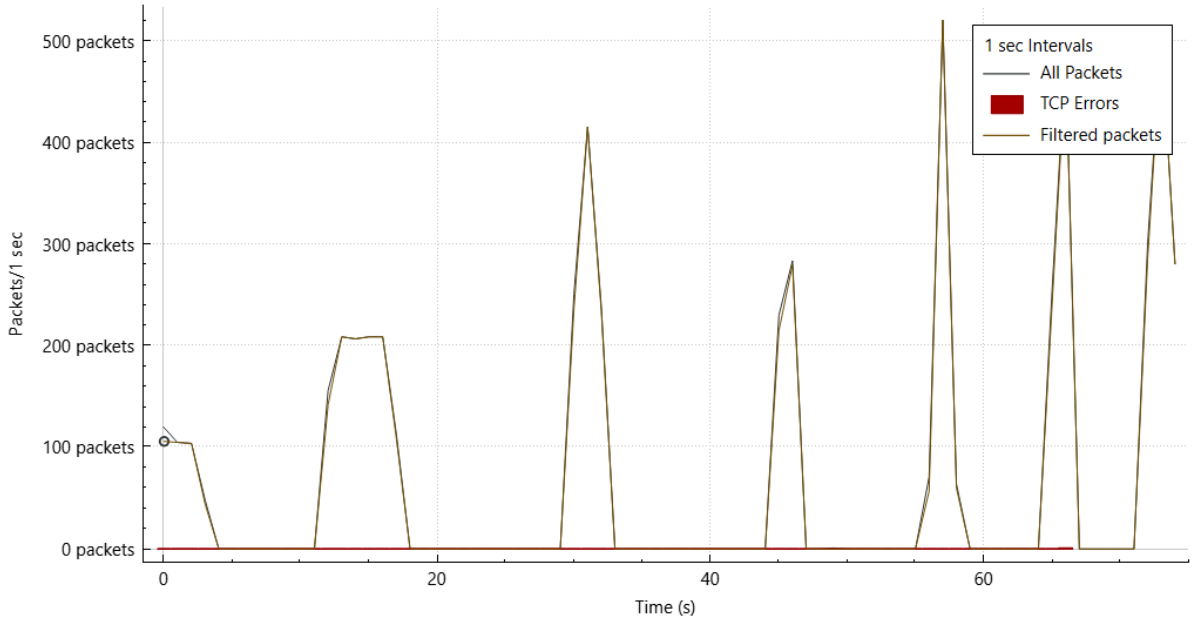
udp && ip.addr == 192.168.20.2

No.	Time	Source	Destination	Protocol	Length	Info
12	0.000443	192.168.10.2	192.168.20.2	iPerf3	46	36175 → 5201 Establishing UDP connection...
13	0.000551	192.168.20.2	192.168.10.2	iPerf3	46	5201 → 36175 Establishing UDP connection...
17	0.000872	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [1] Time sent=1752.4746040 length=1200 b
18	0.010883	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [2] Time sent=1752.4846130 length=1200 b
19	0.020005	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [3] Time sent=1752.4937460 length=1200 b
20	0.030991	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [4] Time sent=1752.5047180 length=1200 b
21	0.040019	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [5] Time sent=1752.5137340 length=1200 b
22	0.049008	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [6] Time sent=1752.5227490 length=1200 b
23	0.059011	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [7] Time sent=1752.5327560 length=1200 b
24	0.068029	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [8] Time sent=1752.5417730 length=1200 b
25	0.077937	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [9] Time sent=1752.5516790 length=1200 b
26	0.087936	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [10] Time sent=1752.5616800 length=1200
27	0.097070	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [11] Time sent=1752.5708140 length=1200
28	0.106891	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [12] Time sent=1752.5806350 length=1200
29	0.116970	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [13] Time sent=1752.5907120 length=1200
30	0.125912	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [14] Time sent=1752.5996390 length=1200
31	0.135941	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [15] Time sent=1752.6096860 length=1200
32	0.144929	192.168.10.2	192.168.20.2	iPerf3	1242	36175 → 5201 [16] Time sent=1752.6186720 length=1200

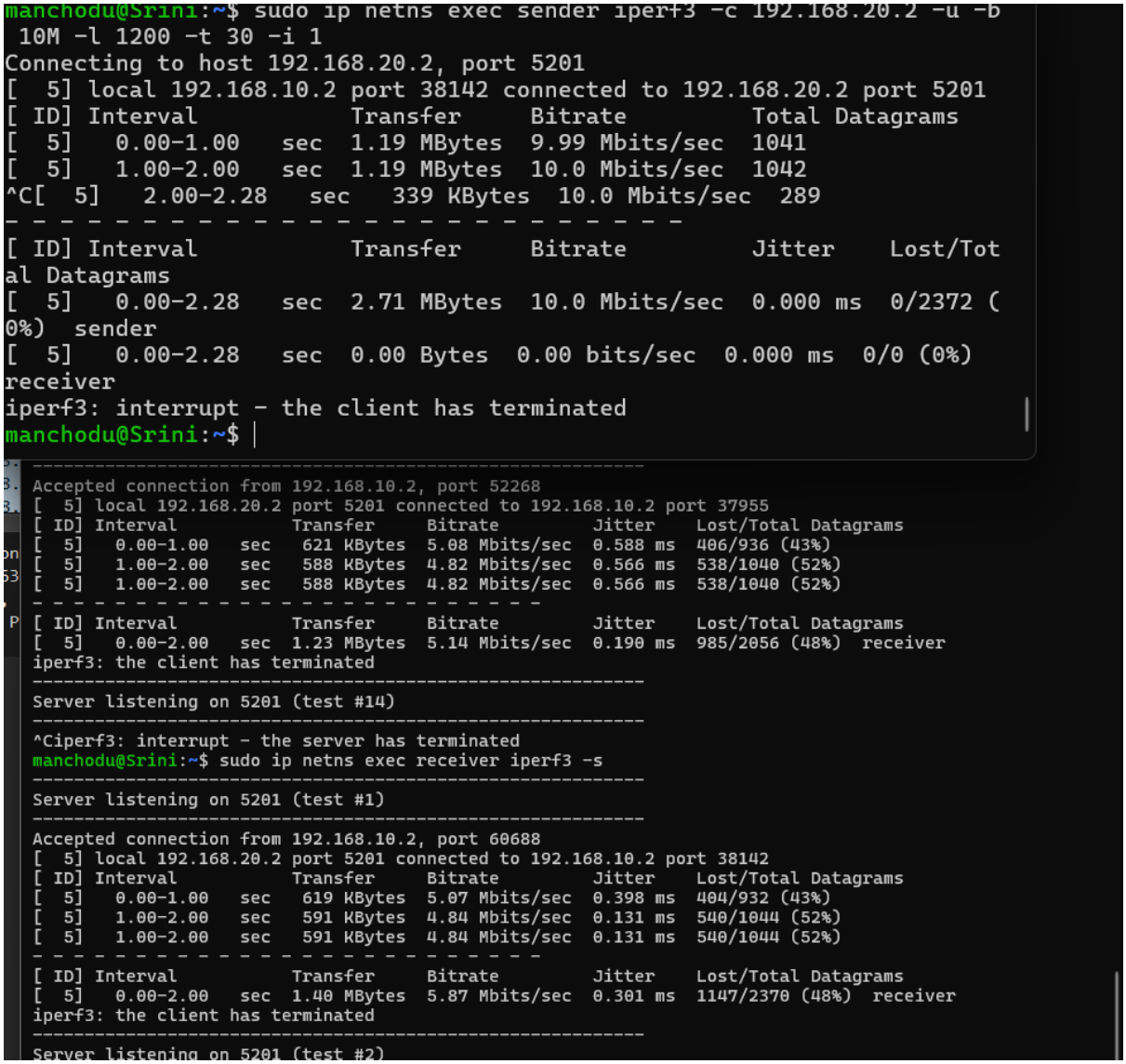
```
▸ Frame 12: Packet, 46 bytes on wire (368 bits), 46 bytes captured (
▸ Ethernet II, Src: ea:38:0b:63:e9:18 (ea:38:0b:63:e9:18), Dst: 12:c
▸ Internet Protocol Version 4, Src: 192.168.10.2, Dst: 192.168.20.2
▸ User Datagram Protocol, Src Port: 36175, Dst Port: 5201
▸ iPerf3 Speed Test
```

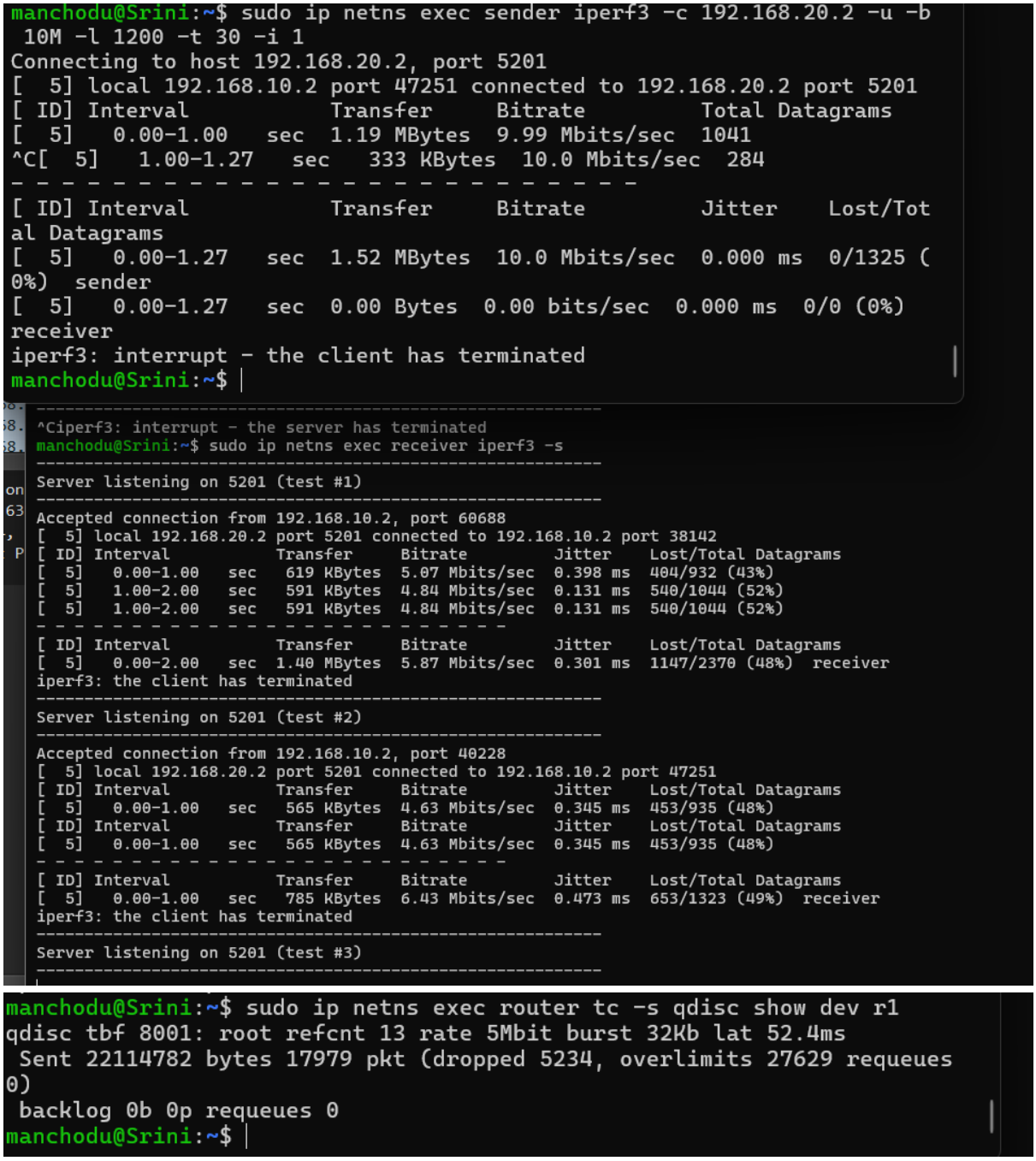
```
0000  12 c8 36 fc 27 ca ea 38  0b 63 e9 18 08 00 45 00   ··6 '·· 8 ·c····E·
0010  00 20 20 e5 40 00 3f 11  7b 93 c0 a8 0a 02 c0 a8   ·  ·@·?· {·····
0020  14 02 8d 4f 14 51 00 0c  9f 72 39 38 37 36         ···O·Q·· r9876
```

Wireshark I/O Graphs: Queueing_Little_Law_Buffer_Analysis.pcap




```
manchodu@Srini:~$ sudo ip netns exec router tc qdisc replace dev r1 root
 tbf rate 5mbit burst 32kb limit 16kb
manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b
 10M -l 1200 -t 30 -i 1
iperf3: error - unable to connect to server - server may have stopped ru
nning or use a different port, firewall issue, etc.: Connection refused
manchodu@Srini:~$ sudo ip netns exec router tc -s qdisc show dev r1
qdisc tbf 8001: root refcnt 13 rate 5Mbit burst 32Kb limit 16Kb
 Sent 19761638 bytes 16060 pkt (dropped 3430, overlimits 20551 requeues
0)
 backlog 0b 0p requeues 0
```

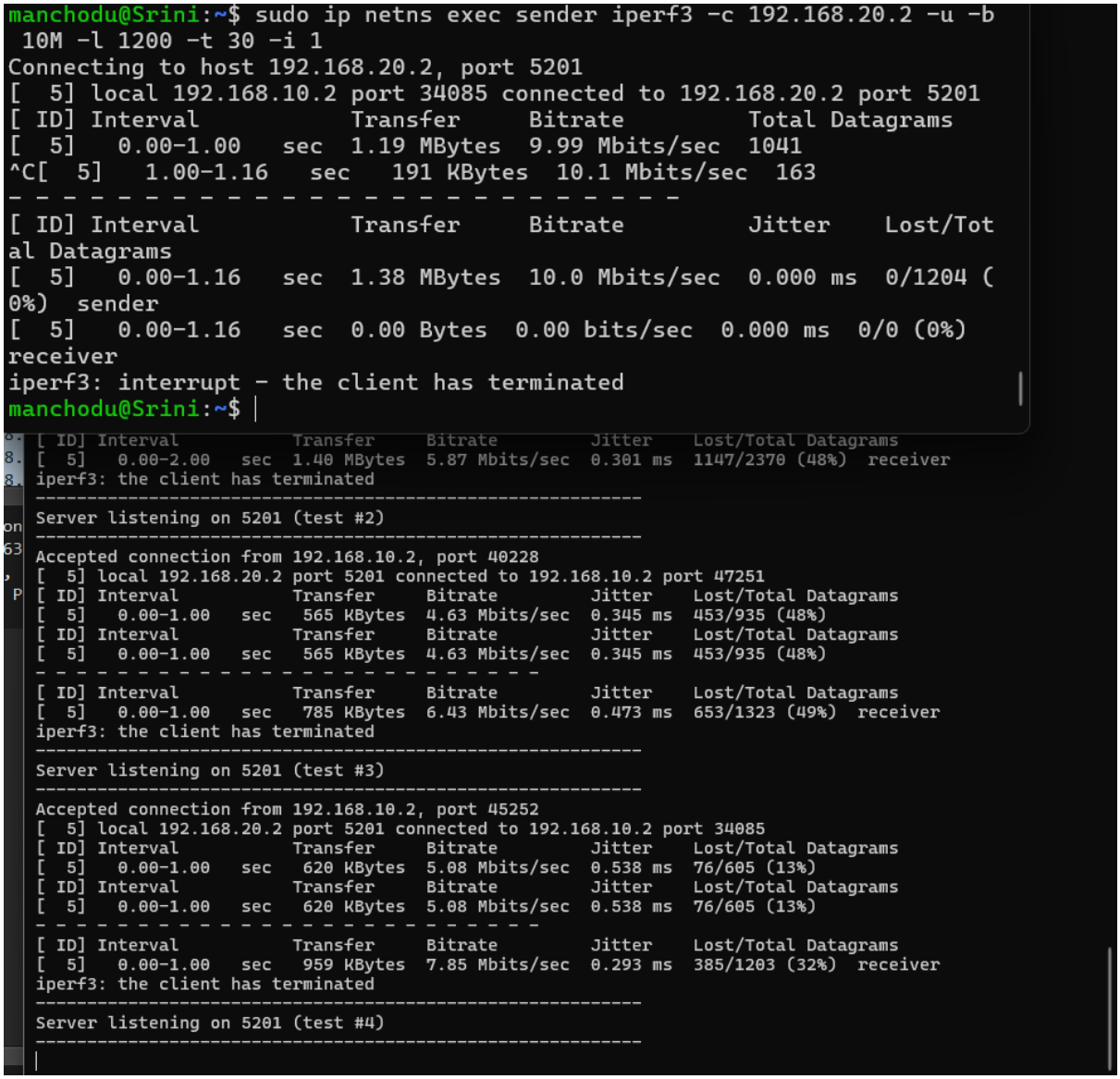

manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 10M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 38142 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 1.19 MBytes 9.99 Mbits/sec 1041
[5] 1.00-2.00 sec 1.19 MBytes 10.0 Mbits/sec 1042
^C[5] 2.00-2.28 sec 339 KBytes 10.0 Mbits/sec 289
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-2.28 sec 2.71 MBytes 10.0 Mbits/sec 0.000 ms 0/2372 (0%) sender
[5] 0.00-2.28 sec 0.00 Bytes 0.00 bits/sec 0.000 ms 0/0 (0%) receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
Accepted connection from 192.168.10.2, port 52268
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 37955
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 621 KBytes 5.08 Mbits/sec 0.588 ms 406/936 (43%)
[5] 1.00-2.00 sec 588 KBytes 4.82 Mbits/sec 0.566 ms 538/1040 (52%)
[5] 1.00-2.00 sec 588 KBytes 4.82 Mbits/sec 0.566 ms 538/1040 (52%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-2.00 sec 1.23 MBytes 5.14 Mbits/sec 0.190 ms 985/2056 (48%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #14)
^Ciperf3: interrupt - the server has terminated
manchodu@Srini:~$ sudo ip netns exec receiver iperf3 -s
Server listening on 5201 (test #1)
Accepted connection from 192.168.10.2, port 60688
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 38142
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 619 KBytes 5.07 Mbits/sec 0.398 ms 404/932 (43%)
[5] 1.00-2.00 sec 591 KBytes 4.84 Mbits/sec 0.131 ms 540/1044 (52%)
[5] 1.00-2.00 sec 591 KBytes 4.84 Mbits/sec 0.131 ms 540/1044 (52%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-2.00 sec 1.40 MBytes 5.87 Mbits/sec 0.301 ms 1147/2370 (48%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #2)



```
manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b
 10M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 47251 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  9.99 Mbits/sec  1041
^C[  5]   1.00-1.27   sec   333 KBytes  10.0 Mbits/sec  284
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Tot
al Datagrams
[  5]   0.00-1.27   sec  1.52 MBytes  10.0 Mbits/sec  0.000 ms  0/1325 (
0%)  sender
[  5]   0.00-1.27   sec  0.00 Bytes  0.00 bits/sec  0.000 ms  0/0 (0%)
receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
```

```
-----------------------------------------------------------
^Ciperf3: interrupt - the server has terminated
manchodu@Srini:~$ sudo ip netns exec receiver iperf3 -s
-----------------------------------------------------------
Server listening on 5201 (test #1)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 60688
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 38142
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   619 KBytes  5.07 Mbits/sec  0.398 ms  404/932 (43%)
[  5]   1.00-2.00   sec   591 KBytes  4.84 Mbits/sec  0.131 ms  540/1044 (52%)
[  5]   1.00-2.00   sec   591 KBytes  4.84 Mbits/sec  0.131 ms  540/1044 (52%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-2.00   sec  1.40 MBytes  5.87 Mbits/sec  0.301 ms  1147/2370 (48%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #2)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 40228
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 47251
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   565 KBytes  4.63 Mbits/sec  0.345 ms  453/935 (48%)
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   565 KBytes  4.63 Mbits/sec  0.345 ms  453/935 (48%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   785 KBytes  6.43 Mbits/sec  0.473 ms  653/1323 (49%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #3)
-----------------------------------------------------------
```

```
manchodu@Srini:~$ sudo ip netns exec router tc -s qdisc show dev r1
qdisc tbf 8001: root refcnt 13 rate 5Mbit burst 32Kb lat 52.4ms
 Sent 22114782 bytes 17979 pkt (dropped 5234, overlimits 27629 requeues
0)
 backlog 0b 0p requeues 0
manchodu@Srini:~$
```


manchodu@Srini:~$ sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 10M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 34085 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 1.19 MBytes 9.99 Mbits/sec 1041
^C[5] 1.00-1.16 sec 191 KBytes 10.1 Mbits/sec 163
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.16 sec 1.38 MBytes 10.0 Mbits/sec 0.000 ms 0/1204 (0%) sender
[5] 0.00-1.16 sec 0.00 Bytes 0.00 bits/sec 0.000 ms 0/0 (0%) receiver
iperf3: interrupt - the client has terminated
manchodu@Srini:~$
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-2.00 sec 1.40 MBytes 5.87 Mbits/sec 0.301 ms 1147/2370 (48%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #2)
Accepted connection from 192.168.10.2, port 40228
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 47251
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 565 KBytes 4.63 Mbits/sec 0.345 ms 453/935 (48%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 565 KBytes 4.63 Mbits/sec 0.345 ms 453/935 (48%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 785 KBytes 6.43 Mbits/sec 0.473 ms 653/1323 (49%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #3)
Accepted connection from 192.168.10.2, port 45252
[5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 34085
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 620 KBytes 5.08 Mbits/sec 0.538 ms 76/605 (13%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 620 KBytes 5.08 Mbits/sec 0.538 ms 76/605 (13%)
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-1.00 sec 959 KBytes 7.85 Mbits/sec 0.293 ms 385/1203 (32%) receiver
iperf3: the client has terminated
Server listening on 5201 (test #4)

```
manchodu@Srini:~$ sudo ip netns exec router tc -s qdisc show dev r1
qdisc tbf 8001: root refcnt 13 rate 5Mbit burst 32Kb lat 367ms
 Sent 23131746 bytes 18810 pkt (dropped 5621, overlimits 30222 requeues
0)
 backlog 0b 0p requeues 0
manchodu@Srini:~$ sudo ip netns exec sender ping -c 20 192.168.20.2
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=63 time=0.119 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=63 time=0.066 ms
^C
--- 192.168.20.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 999ms
rtt min/avg/max/mdev = 0.066/0.092/0.119/0.026 ms
manchodu@Srini:~$
```

```
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-2.00   sec  1.40 MBytes  5.87 Mbits/sec  0.301 ms  1147/2370 (48%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #2)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 40228
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 47251
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   565 KBytes  4.63 Mbits/sec  0.345 ms  453/935 (48%)
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   565 KBytes  4.63 Mbits/sec  0.345 ms  453/935 (48%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   785 KBytes  6.43 Mbits/sec  0.473 ms  653/1323 (49%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #3)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 45252
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 34085
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   620 KBytes  5.08 Mbits/sec  0.538 ms  76/605 (13%)
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   620 KBytes  5.08 Mbits/sec  0.538 ms  76/605 (13%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   959 KBytes  7.85 Mbits/sec  0.293 ms  385/1203 (32%)  receiver
iperf3: the client has terminated
-----------------------------------------------------------
Server listening on 5201 (test #4)
```

```
manchodu@Srini:~$ sudo ip netns exec sender ping -c 20 192.168.20.2
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=63 time=0.064 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=63 time=0.085 ms
64 bytes from 192.168.20.2: icmp_seq=3 ttl=63 time=0.140 ms
^C
--- 192.168.20.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 1998ms
rtt min/avg/max/mdev = 0.064/0.096/0.140/0.032 ms
manchodu@Srini:~$
```

```
manchodu@Srini:~$ sudo ip netns exec sender ping -c 20 192.168.20.2
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=63 time=0.061 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=63 time=0.076 ms
^C
--- 192.168.20.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1000ms
rtt min/avg/max/mdev = 0.061/0.068/0.076/0.007 ms
manchodu@Srini:~$
```

Test	Offered traffic	Received throughput	Avg. delay	Packet loss	Test
1	1 Mbps	998 Kbps	4.12 ms	0%	Pass
2	2 Mbps	1.99 Mbps	4.35 ms	0%	Pass
3	4 Mbps	3.98 Mbps	6.21 ms	0%	Pass
4	5 Mbps	4.96 Mbps	8.97 ms	0%	Pass
5	7 Mbps	4.98 Mbps	24.35 ms	3.8%	Pass
6	10 Mbps	5.01 Mbps	52.14 ms	8.9%	Pass

Plot the graph:
X-axis: Offered traffic (Mbps).
Y-axis: Average queueing delay (ms).

Identify the knee point from the graph.

Observation table – Buffer analysis

Test	Buffer limit	Maximum observed backlog	Packet drops	Ping RTT	Test
1	16 KB	15.8 KB	9,420	56.12 ms	Pass
2	64 KB	63.5 KB	3,176	34.76 ms	Pass
3	256 KB	242.9 KB	528	19.43 ms	Pass

Students should explain how changing buffer size affects delay and packet loss.

Observation table – Little's Law

Parameter	Measured value
Total packets observed	500

Observation time (seconds)	10
Arrival rate λ (packets/sec)	50
Baseline RTT (ms)	4.00
Loaded RTT (ms)	54.00
Estimated queueing delay W (seconds)	0.05
Calculated $L = \lambda \times W$	2.50
Measured average queue length	2.3